100kW Grid Tied Petrolium Exploration Limited G8 petrolium exploration limited g8

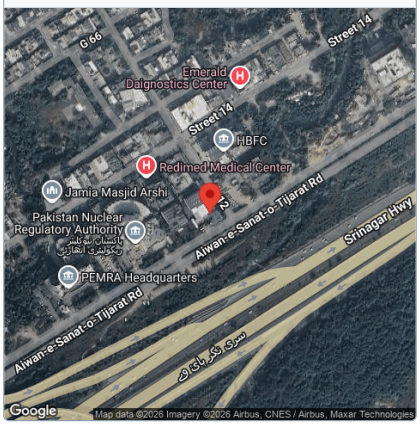
Project Details

Address	petrolium exploration limited g8
Owner	Usama Muhammad
Last Modified	Usama Muhammad 16 minutes ago
Location	(33.68765339999999, 73.0509285) (GMT 5)
AC Interconnect	400Y/230V
Profile	Default Commercial

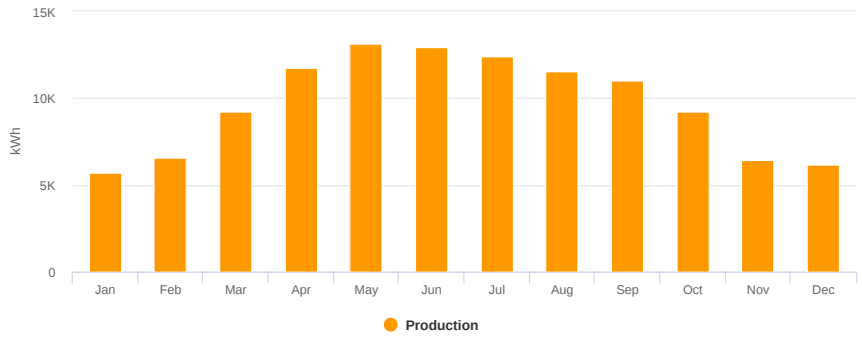
System Metrics

Design	100kW Grid Tied
Module DC Nameplate	88.16 kW
Inverter AC Nameplate	100.00 kW Load Ratio: 0.88
Annual Production	116.2 MWh
Performance Ratio	80.4%
kWh/kWp	1,317.5
Weather Dataset	TMY, 10km Grid, Meteonorm 8 (meteonorm_v8)
Simulator Version	7bb23cb566-43e555984b-8b162cc70b-efbefe9c7c

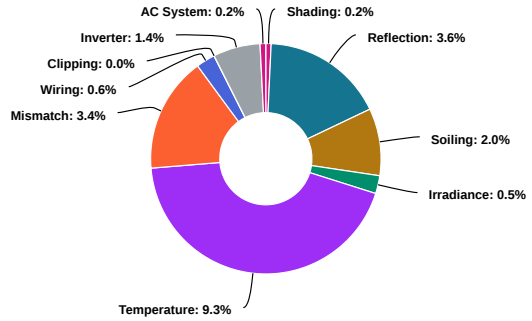
Project Location



Energy Consumption



System Loss



Annual Production

	Description	Output	% Delta
Irradiance (kWh/m²)	Annual Global Horizontal Irradiance	1,576.5	-
	Adjusted Global Horizontal Irradiance	1,576.5	-0.0%
	POA Irradiance	1,638.3	3.9%
	Shaded Irradiance	1,635.7	-0.2%
	Irradiance After Reflection	1,576.1	-3.6%
	Irradiance After Soiling	1,544.6	-2.0%
	Total Collector Irradiance	**1,544.6**	**-0.0%**
Energy (kWh)	Nameplate	136,188.4	-
	Output at Irradiance Levels	135,467.1	-0.5%
	Output at Cell Temperature Derate	122,893.2	-9.3%
	Output after Electrical Mismatch	118,671.9	-3.4%
	Optimal DC Output	118,000.5	-0.6%
	Constrained DC Output	118,000.3	-0.0%
	Inverter Output	116,348.3	-1.4%
	AC System Output	116,150.9	-0.2%
	Energy to Grid	**116,150.9**	**0.0%**

Temperature Metrics	
Avg. Operating Ambient Temp	25.5°C
Avg. Operating Cell Temp	42.4°C

Simulation Metrics	
Operating Hours	4,602

Condition Set

Description	Condition Set 1
Weather Dataset	TMY, 10km Grid, Meteonorm 8(meteonorm_v8)
Solar Angle Location	Meteo Lat/Lng
Transposition Model	Perez Model
Temperature Model	Sandia Model

Temperature Model Parameters

Rack Type	a	b	Temperature Delta
Fixed Tilt	-3.56	-0.08	3.0°C
Flush Mount	-2.81	-0.05	0.0°C
East-West	-3.56	-0.08	3.0°C
Carport	-3.56	-0.08	3.0°C

Soiling (%)

J	F	M	A	M	J	J	A	S	O	N	D
2	2	2	2	2	2	2	2	2	2	2	2

Albedo

J	F	M	A	M	J	J	A	S	O	N	D
0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2

Rear Mismatch Loss	10%	**Rear Shading Factor**	5%
Module Transparency	0%		
Irradiation Variance	5.0%		
Cell Temperature Spread	4.0°C		
Module Binning Range	-2.5% to 2.5%		
AC System Derate	0.50%		

Component Characterizations

Type	Component	Characterization	Bifacial
Module	CS6W-580T (1000V) (Canadian Solar)	Spec Sheet Characterization, PAN	False
Inverter	SUN2000-100KTL-M1 (380/400) (Huawei)	Spec Sheet	N/A

Design BOM

Component	Type	Specs
1/0 AWG (Aluminum)	AC Home Runs	1 (170.4 ft)
8 input Combiners	Combiners	1 (-)
10 AWG (Copper)	Home Runs	1 (40.6 ft)
SUN2000-100KTL-M1 (380/400) (Huawei)	Inverters	1 (100.00 kW)
Canadian Solar, CS6W-580T (1000V), (580W)	Modules	152 (88.16 kW)
10 AWG (Copper)	Strings	8 (852.5 ft)

Monthly Shading

Month	GHI (kWh/m^2)	POA (kWh/m^2)	Shaded (kWh/m^2)	Nameplate (kWh)	Grid (kWh)
January	70.6	77.5	77.3	6,370.5	5,725.5
February	82.5	88.6	88.4	7,322.8	6,558.3
March	122.5	127.7	127.6	10,606.0	9,228.3
April	160.9	165.5	165.3	13,812.7	11,728.3
May	187.0	188.8	188.6	15,790.9	13,126.3
June	186.3	186.8	186.5	15,597.9	12,948.0
July	176.8	178.0	177.8	14,844.4	12,427.2
August	161.8	165.1	164.9	13,766.7	11,576.6
September	150.7	157.3	157.1	13,107.8	11,002.2
October	121.3	129.5	129.3	10,737.5	9,206.8
November	80.9	88.8	88.6	7,300.9	6,427.9
December	75.3	84.5	84.3	6,930.3	6,195.4

Design Wiring Zone

Description	Combiner Poles	String Size	Stringing Strategy
Wiring Zone	12	5 - 19	Along Racking

Design Render



Field Segments

Description	Racking	Orientation	Tilt	Azimuth	Intrarow Spacing	Frame Size	Frames	Modules	Power
Field Segment 1	Flush Mount	Portrait (Vertical)	7°	148°	0.1 ft	1x1	96	96	55.68 kW
Field Segment 2	Flush Mount	Landscape (Horizontal)	7°	148°	0.1 ft	1x1	28	28	16.24 kW
Field Segment 3	Flush Mount	Landscape (Horizontal)	7°	148°	0.1 ft	1x1	28	28	16.24 kW